

Receive and Check In a Carton SOP

One episode receives one carton. The tape strip is cut along the seam, both flaps are folded down, the packing slip is laid face up on the desk, and the six items come out of the carton into one row across the front of the desk, left to right in slip line order. The carton is empty, both flaps are still down, and the cutter is back on its rest when the episode ends.

The order never changes: cut, flaps, slip, then the six items in slip line order 1 to 6. Nothing is lifted out of the carton until the slip is face up on the desk, and the line number is read before each item is lifted.

The carton sits on the bare desk with nothing holding it, so the gripper that is not working anchors it. The carton is never lifted, slid, tipped, or turned, and no gripper touches the seam side of the carton while the other is cutting there.

The right gripper works the safety cutter, the right flap, and slip lines 4, 5, and 6. The left gripper works the left flap, the slip, and slip lines 1, 2, and 3, and it anchors the carton for the cut. Only the right gripper ever touches the cutter.

The blade only ever travels along the tape strip, no flap is torn open against tape that is still joined, nothing is turned in the air, no item is pulled out from between its neighbors, and no gripper opens anywhere but over the spot the item is going into.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole desk: the carton at back center, the clear band across the front where the row is built, the slip spot at the front left, and the cutter rest at the front right.
3. The whole tape seam is in frame end to end, and the carton interior is visible from above so every lift can be seen leaving it.
4. Both arms are at home with grippers open.
5. The desk is bare apart from the carton and the cutter. There are no marks, trays, bins, or fixtures on it.
6. Nothing obstructs the paths between the carton and the front of the desk. The left arm reaches the carton's left wall, the carton's left half, the left part of the row, and the slip spot. The right arm reaches the carton's right wall, the whole tape seam, the carton's right half, the right part of the row, and the cutter rest.

Materials checklist

1. One carton sitting square on the desk at back center, base flat on the desk, both top flaps closed and meeting along the seam, with clear desk on all four sides of it.
2. The carton has two top flaps only, hinged along the left and right top edges, meeting edge to edge along a **seam** that runs front to back across the top.
3. One **tape strip** lies flat along the seam, covering it end to end, with no tape carried down the front, back, or side walls.
4. One **safety cutter** lies flat on the desk at the front right, head pointing away from the front edge, with a recessed hook blade seated and its depth set so a flat draw parts tape only.
5. One packing slip, printed, stiff enough to keep its shape when pinched by an edge, listing six numbered lines with one item per line. It lies face up on top of the contents in the carton's left half.
6. Six items lie in the carton in one layer, in a single row running left to right in slip line order, with a clear gap between them so a gripper can close on one without touching its neighbors. The six are one **mug box**, one **book**, one **bottle** standing upright with its cap closed, one **cable pouch** lying flat, one **tin** with its lid up, and one **marker box** lying flat. Nothing rests on top of anything else.
7. The front of the desk is clear end to end, with room for six items in a row.

Workspace layout

Every zone is a position on the bare desk. Nothing is marked and nothing holds anything, so each position is held by where the last thing was put.

- **Carton:** at back center, centered between the two arms, close enough that both arms reach its top without leaning out. It sits on the desk for the whole episode.
- **Row band:** the clear band of desk across the front, between the carton and the front edge. The six items are built into one row here, left to right in slip line order.
- **Slip spot:** the front left corner of the desk, left of where the row starts and clear of it.
- **Cutter rest:** the front right corner of the desk, right of where the row ends and clear of it. The cutter lies flat here, head away from the front edge, before and after the cut.

Lines 1 to 3 are packed in the carton's left half and go to the left part of the row. Lines 4 to 6 are packed in the right half and go to the right part. No carried item crosses the midline and no carried item passes over another item.

Vocabulary

- **Arm assignment:** the right arm works the cutter, the tape seam, the right flap, the carton's right half, and the right part of the row. The left arm works the left flap, the slip, the carton's

left half, and the left part of the row. Either arm anchors on the wall on its own side. Neither arm takes the other's zones.

- **Seam:** the line where the two closed flaps meet, running front to back across the top of the carton. It has a **far end** and a **near end**.
- **Tape strip:** the single strip covering the seam. It is the only thing the blade ever touches.
- **Safety cutter:** the tool on the cutter rest. Its **handle** has a flat grip section and its **head** carries the recessed blade. A gripper closes on the handle and never on the head.
- **Anchor:** the closed gripper pressed onto the outside of the carton wall on its own side, level with the middle of the wall, holding the carton still on the bare desk while the other gripper works. An anchor never travels, never sits on the top of the carton, and lifts clear only after the working gripper is off the carton. Nothing else holds the carton, so every cut and every flap fold is anchored.
- **Draw:** the right gripper carrying the cutter along the seam, far end to near end, in one continuous motion with the sole of the head flat on the carton top.
- **Flap down:** a flap folded out over its hinge, creased, and left lying flat against the outside of its own wall, leaving the top opening clear front to back.
- **Slip line:** a numbered line on the packing slip. The item on line N is the Nth item placed in the row, and the line number is read before the item is lifted.
- **Row:** the six items standing on the bare desk in one line across the front, built left to right in slip line order, roughly in line with each other, with a clear gap between neighbors and nothing touching. The row order is the count, so an item out of order is a miscount.
- **Next spot:** the clear desk just to the right of the last item placed, far enough out that the two do not touch. The first item sets where the row starts.
- **Item grasps:** the mug box across its two side walls; the book across its spine edge, taking the full thickness; the bottle around its body, below the cap and above the base, upright; the cable pouch across its middle; the tin around its rim wall, lid up; the marker box across its middle.
- **Lift:** taking an item straight down onto its grasp, closing, and raising it straight up until there is daylight under it, before anything moves sideways.
- **Stage:** carrying an item level to its next spot, lowering it straight down until it is resting flat on the desk, and releasing once it is resting.
- **Settled:** the item stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or falling over.
- **Release point:** over the next spot, the slip spot, or the cutter rest the thing in the gripper is going to, once it is already resting. A gripper opens nowhere else.

Steps

Steps 1 to 5 are one carton. Every item is lifted straight up out of the carton and carried in the orientation it will be set down in.

Step 1: Cut the tape along the seam

Goal: the tape strip parted along the whole seam, the carton still square where it started, and the cutter back on its rest.

- The **left gripper**, closed, anchors the carton on its **left wall** and holds it there for the whole step.
- The **right gripper** comes down onto the flat of the **handle** on the cutter rest, closes across it, and lifts the cutter straight up.
- Bring the head down onto the **far end** of the tape strip until the sole of the head sits flat on the carton top and the blade is through the tape.
- Draw the cutter along the seam, far end to near end, in one continuous motion, keeping the head flat and on the strip, until the blade passes the near end of the tape.
- Lift the cutter straight up clear of the carton, carry it back to the **cutter rest**, lower it flat onto the desk head away from the front edge, and release.
- The **left gripper** lifts clear once the cutter is on its rest.

Nothing but the anchor holds the carton, so the draw is pulled toward the front edge against a gripper pressed on the opposite wall. The blade is set to tape depth and the head stays flat, so the cut parts the tape without reaching the contents. Never lift the head off the strip and set it down again partway along the seam.

Check: the tape is parted along the whole seam, the two flaps are free of each other, the carton sits square where it started and has not slid or turned, and the cutter lies on its rest. If the tape is still joined anywhere, anchor again and draw the joined part with the same flat motion. Never pull the flaps apart against tape that is still joined.

Step 2: Fold both flaps down

Goal: both flaps folded down against their own outside walls with the top opening clear.

- The **left gripper** anchors the carton on its **left wall**. The **right gripper** pinches the free edge of the **right flap** at its middle, lifts it up and out over its hinge in one continuous motion, lays it flat against the carton's right outside wall, presses it there for 2 seconds so the crease sets, and releases.
- Swap the roles: the **right gripper** anchors the carton on its **right wall**, and the **left gripper** takes the **left flap** the same way onto the carton's left outside wall.
- Both grippers lift clear.

Each flap is opened by the gripper on its own side and anchored by the other, so the fold pulls against a held wall instead of dragging the carton across the desk.

Check: both flaps lie down against their own outside walls, neither one is standing up or leaning back over the opening, the top opening is clear front to back, the packing slip is in view, and the carton has not moved. If a flap springs back over the opening, anchor again, take its free edge, and crease it down again.

Step 3: Lay the packing slip on the desk

Goal: the slip lying face up at the slip spot and readable, with all six items uncovered.

- The **left gripper** pinches the near edge of the slip at its middle and lifts it straight up out of the carton, level and face up.
- Carry it level to the **slip spot** at the front left, lower it flat onto the desk square to the front edge, and release once it is resting.
- The slip stays there untouched for the rest of the episode.

Check: the slip lies flat and face up at the front left, its six lines readable in frame, clear of the band where the row will be built, and all six items in the carton are uncovered. If the slip lands face down, folded over, or in the way of the row, lift it clear and lay it again.

Step 4: Stage lines 1, 2, and 3

Goal: the carton's left half empty and the first three items standing in the row in slip line order.

Read the line on the slip, then take the item named on it.

- The **left gripper** closes on the **line 1** item at its grasp, lifts it straight up out of the carton, carries it level to the left end of the **row band**, lowers it until it is resting flat on the desk, and releases. This item sets where the row starts.
- Stage the **line 2** item at the **next spot** to its right, and the **line 3** item at the next spot after that, each one lowered straight down and left with a clear gap to its neighbor.
- Move to the next line only once the current item is resting and settled.

Take each item straight up. Never slide one out from between its neighbors, never move one item aside to reach another, and never nudge a placed item to make room for the next one. Leave the gap when you set the item down.

Check: three items stand across the left of the row in slip line order, each resting flat, settled, roughly in line with the others, with a clear gap between neighbors and none touching, the bottle upright if it is one of these three, and the carton's left half is empty. If an item is out of order, lift it clear and lower it in its right place.

Step 5: Stage lines 4, 5, and 6

Goal: the carton empty and all six items standing in one row in slip line order.

- The **right gripper** stages the **line 4** item at the next spot to the right of item 3, then the **line 5** item, then the **line 6** item, each one read from the slip first and each one taken at its grasp, lifted straight up, carried level, and lowered until it is resting flat on the desk.
- Move to the next line only once the current item is resting and settled.

Never tip or tilt the carton to bring an item within reach. If an item sits deep in the carton, come down on it from above.

Check: six items stand in one row across the front in slip line order left to right, each resting flat and settled, roughly in line with each other, with a clear gap between neighbors and none touching, the tin lid up and the bottle upright, and the carton floor is bare from front to back.

Step 6: Check the count and end the episode

- Read the slip line by line and confirm each of the six lines has its item standing in the row, in the same order the slip lists them.
- Confirm the carton is empty, both flaps are still down, and the cutter lies on its rest.
- Return both arms home, clear of the carton and the row, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the six items off the desk and lay them back in the carton in one layer, in a single row running left to right, with a clear gap between them. The bottle stands upright with its cap closed and the tin sits lid up. Nothing rests on top of anything else.
2. Vary which item sits on which line between episodes rather than rebuilding the same order every time, and print a slip whose six lines match the row the items were packed in.
3. Lay the slip face up on top of the contents in the carton's left half.
4. Lift both flaps off the outside walls and fold them back over the opening so they meet edge to edge along the seam. Flatten the creases so neither flap stands proud.
5. Lay a fresh tape strip flat along the seam, covering it end to end, with no tape carried down the walls. Clear the cut strip off the desk.
6. Set the carton square at back center, base flat on the desk, flaps closed and facing up, with clear desk on all four sides.
7. Check the cutter: blade seated, depth set to tape only, and the head parts a strip in one flat draw. A blade that drags, tears, or leaves the tape joined is replaced before the next episode. Lay it on the cutter rest head away from the front edge.
8. Replace any item that is damaged: a carton that will not sit square or whose flaps will not stay creased down, a crushed box, a leaking or capless bottle, a tin whose lid has come off, or a slip that is torn or no longer readable.

9. Wipe the desk and confirm it is dry and bare. A slick or crumbed desk lets the carton slide under an anchor.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a flap is opened before the tape is parted along the whole seam, the slip is lifted before both flaps are down, an item is lifted before the slip is resting on the desk, or the lines are staged in any order other than 1 to 6.
- **SOP rule broken:** Steps 1 to 5, the order never changes: cut, flaps, slip, then the six items in slip line order 1 to 6.
- **Coaching note:** finish the step you are in before you start the next one.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes the cutter, the right flap, or a line 4 to 6 item; or the right gripper takes the left flap, takes the slip, or reaches the left part of the row.
- **SOP rule broken:** Steps 1 to 5, each arm keeps its own tool, flap, half of the carton, and part of the row.
- **Coaching note:** each arm stays on its own side. Nothing crosses the middle.

Violation: Worked without an anchor

- **Visible cue:** the cutter is on the seam or a flap is being folded while the other gripper is off the carton, or the anchoring gripper lets go before the working gripper is clear.
- **SOP rule broken:** Steps 1 and 2, nothing else holds the carton, so every cut and every flap fold is anchored on the opposite wall.
- **Coaching note:** anchor first, let go last.

Violation: Anchor in the wrong place

- **Visible cue:** the anchoring gripper presses on the top of the carton, on a flap, on the seam side, or on the wall the other gripper is working, or it slides along the wall during the step.
- **SOP rule broken:** Steps 1 and 2, the anchor sits still on the outside of the wall on its own side, never on the top.
- **Coaching note:** flat on the side wall, and hold it still.

Violation: Carton moved

- **Visible cue:** the carton slides, turns, tips, or lifts off the desk at any point, or it is dragged toward the front edge to bring the contents within reach.
- **SOP rule broken:** Steps 1 to 5, the carton is never lifted, slid, tipped, or turned.
- **Coaching note:** if the carton is moving, the anchor is not holding. Stop, reseal it, and anchor again.

Violation: Cutter taken by the head

- **Visible cue:** the gripper closes on the head of the cutter, on the blade end, or across the cutter at an angle instead of on the flat of the handle.
- **SOP rule broken:** Step 1, a gripper closes on the flat of the handle only, never on the head.
- **Coaching note:** land on the handle or reset and take it again.

Violation: Blade off the tape strip

- **Visible cue:** the cutter wanders off the seam onto the flap face, is drawn in an arc rather than along the line of the seam, is drawn near end to far end, or scores the carton beside the strip.
- **SOP rule broken:** Step 1, one continuous motion along the seam, far end to near end, and the blade touches nothing but the tape.
- **Coaching note:** put the head on the strip and keep it there for the whole run.

Violation: Head lifted or plunged during the cut

- **Visible cue:** the head is lifted off the strip partway along and set down again, or the cutter is driven down into the carton rather than kept flat, and the blade reaches below the tape.
- **SOP rule broken:** Step 1, keep the sole of the head flat on the carton top for the whole draw.
- **Coaching note:** flat on, flat all the way, flat off.

Violation: Flaps forced against joined tape

- **Visible cue:** a flap is pulled, levered, or torn open while part of the tape strip is still joined, or the tape rips rather than parting along the cut.
- **SOP rule broken:** Steps 1 and 2, no flap is torn open against tape that is still joined.
- **Coaching note:** if it resists, put the cutter back on the joined part.

Violation: Cutter not returned to its rest

- **Visible cue:** the cutter is left on the carton, in the row band, on the slip, or still in the gripper when the flaps are opened, or it is dropped toward the desk rather than lowered onto it.
- **SOP rule broken:** Step 1, the cutter is back flat on the cutter rest before any flap is touched.
- **Coaching note:** the tool goes back before the next step starts.

Violation: Flap left over the opening

- **Visible cue:** a flap is left standing up, leaning back across the opening, or resting on the contents, and items are lifted out past it.
- **SOP rule broken:** Step 2, both flaps lie creased down against their own outside walls with the opening clear front to back.
- **Coaching note:** crease it down and hold it before you let go.

Violation: Slip mishandled

- **Visible cue:** the slip is laid face down, lands in the row band or under an item, is folded or creased, is left in the carton, or is moved, covered, or pushed after it has been laid down.
- **SOP rule broken:** Step 3, the slip lies face up and readable at the front left and stays untouched for the rest of the episode.
- **Coaching note:** face up, out of the way, then leave it alone.

Violation: Row out of slip order

- **Visible cue:** an item lands out of its slip line position, an item is placed to the left of one already down, or the row reads in an order the slip does not list.
- **SOP rule broken:** Steps 4 and 5, the row is built left to right in slip line order and the row order is the count.
- **Coaching note:** read the line, name the item, then put it at the right hand end of the row.

Violation: Items crowded or touching in the row

- **Visible cue:** two staged items touch, lean on each other, or sit stacked, an item is set down so close that the next one has nowhere to go, or the row is left as a cluster rather than a line.
- **SOP rule broken:** Steps 4 and 5, a clear gap between neighbors and nothing touching.
- **Coaching note:** leave the gap when you set it down, not afterwards.

Violation: Placed item moved to make room

- **Visible cue:** a gripper nudges, slides, or lifts an item that is already in the row to fit the next one in, or the row is shuffled along the desk.

- **SOP rule broken:** Steps 4 and 5, never nudge a placed item to make room for the next one.
- **Coaching note:** the row grows to the right. Nothing already down gets touched.

Violation: Item dug out from between its neighbors

- **Visible cue:** an item is slid, pulled, or worked out from between the items beside it, or a gripper moves one item aside to reach another.
- **SOP rule broken:** Steps 4 and 5, no item is pulled out from between its neighbors and each item is taken straight up.
- **Coaching note:** everything is in one layer already. Come down on it and lift straight up.

Violation: Item taken by the wrong part

- **Visible cue:** the bottle is taken by its cap, the tin by its lid, the book by a single cover or a page block, the cable pouch by a corner, a box by an edge or a flap, or the slip by a corner rather than its near edge.
- **SOP rule broken:** Steps 3 to 5, each item has one grasp.
- **Coaching note:** land on the grasp point or reset and take it again.

Violation: Item dragged instead of lifted

- **Visible cue:** an item slides along the carton floor, the carton wall, or the desk still touching it, with no daylight under it in transit.
- **SOP rule broken:** Steps 3 to 5, lift straight up until there is daylight before moving sideways.
- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Item turned in the air

- **Visible cue:** the bottle tips off vertical, the tin rolls off its lid up carry, a box or the book is flipped or swung during the carry, or the slip is turned over between the carton and the desk.
- **SOP rule broken:** Steps 3 to 5, nothing is turned in the air and each item is carried in the orientation it will be set down in.
- **Coaching note:** orientation is set at the pick, never in the air.

Violation: Regripped in the air

- **Visible cue:** the gripper shifts, rolls, or seats its hold again on an item or on the cutter without setting it down first.
- **SOP rule broken:** Steps 1 to 5, set it down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Item dropped instead of lowered

- **Visible cue:** a gripper opens above the desk and the item falls the rest of the way, or an item is set down short of its spot and slid across the desk into place rather than lowered straight down onto it.
- **SOP rule broken:** Steps 1 to 5, release only once the item is resting.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the carton, over the slip, over another item, or over open space, and the item lands anywhere but at the spot it was going to.
- **SOP rule broken:** Steps 1 to 5, a gripper opens over the spot the item is going to and nowhere else.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Contents pressed or packed down

- **Visible cue:** a gripper presses, pats, or squashes an item in the row, the slip on the desk, or the items still in the carton, or holds an item down against the desk after it is already resting.
- **SOP rule broken:** Steps 4 and 5, each item is lowered until it is resting and nothing is pressed down. The 2 second press in Step 2 is for the flap crease only.
- **Coaching note:** leave it how it lands.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides an item into position after releasing it, instead of lifting it clear and lowering it again.
- **SOP rule broken:** Steps 3 to 5, correct a bad placement by lifting the item clear and lowering it into place again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** an item or the cutter falls onto the desk short of its spot or onto the floor, the carton is struck and shifts, an already staged item is knocked over or out of the row, the bottle is knocked over, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 5, keep each item on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with something still in the carton, an item missing from the row, the row out of slip order, a flap back over the opening, the cutter off its rest, an arm away from home, an arm still over the carton or the row.

- **SOP rule broken:** Step 6, confirm the count and the empty carton; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm, hold, then home. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Blade will not part the tape** under a flat draw along the strip, or the blade is missing, dull, or set to the wrong depth. Replace the cutter before the next episode.
- **Tape strip missing, already split, or wrapped down the walls** when the episode begins.
- **Packing slip missing, blank, or unreadable** when the step begins.
- **Count mismatch:** the carton holds fewer or more items than the slip lists, or an item in the carton is not on the slip. Discard the episode and restock the carton to match the slip.
- **Carton slides under a correct anchor** because the desk is slick, wet, or crumbed. Wipe and dry the desk before the next episode.
- **Carton will not sit square** on the desk, or a flap will not hold its crease and springs back over the opening under its own weight. Replace the carton before the next episode.
- **Defective item:** a leaking or capless bottle, a crushed box, a tin whose lid has come off, a pouch that has split open, or an item that will not stand or lie still on the desk. Replace it before the next episode.

Annotation subtasks (from SOP)

1. Anchor the carton on a side wall
2. Take the safety cutter from its rest
3. Draw the cutter along the tape seam
4. Return the safety cutter to its rest
5. Fold one flap down against its outside wall
6. Lay the packing slip on the desk
7. Move one item from the carton into the row
8. Confirm the count and end the episode